Physics for Scientists and Engineers Storyline

Tenth Edition

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Steady state flow of charges

Chapter 26

Current and Resistance

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Electric Current (1 of 2)



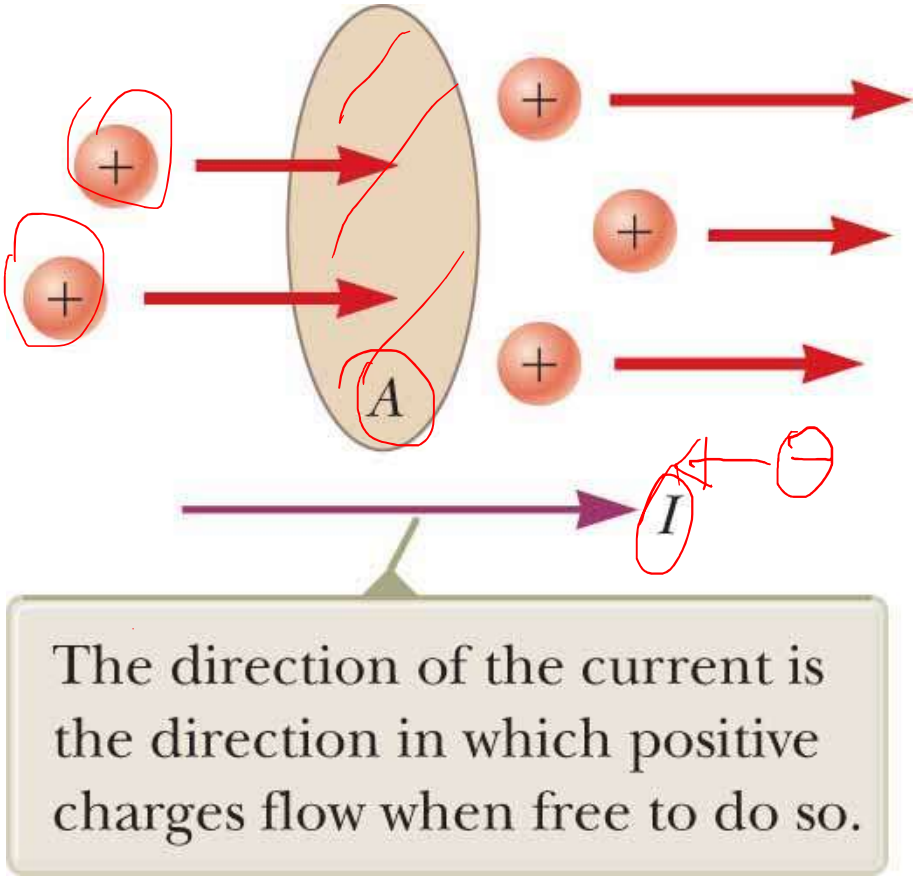
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Electric Current (2 of 2)

$$I_{\text{avg}} = \frac{\Delta Q}{\Delta t}$$

$$I \equiv \frac{dQ}{dt}$$

$$1\ \text{A} = 1\ \text{C/s}$$



The direction of the current is the direction in which positive charges flow when free to do so.

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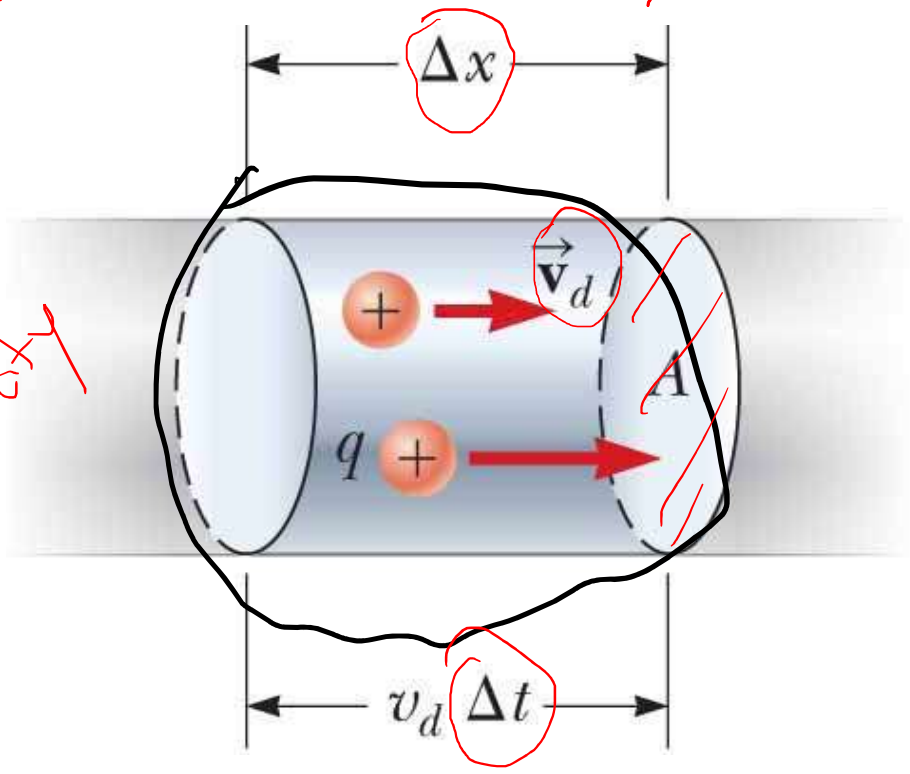
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Microscopic Model of Current (1 of 2)

$$\Delta Q = (nA\Delta x)q$$

$$\frac{\Delta Q}{\Delta t} = (nAv_d\,\Delta t)q$$

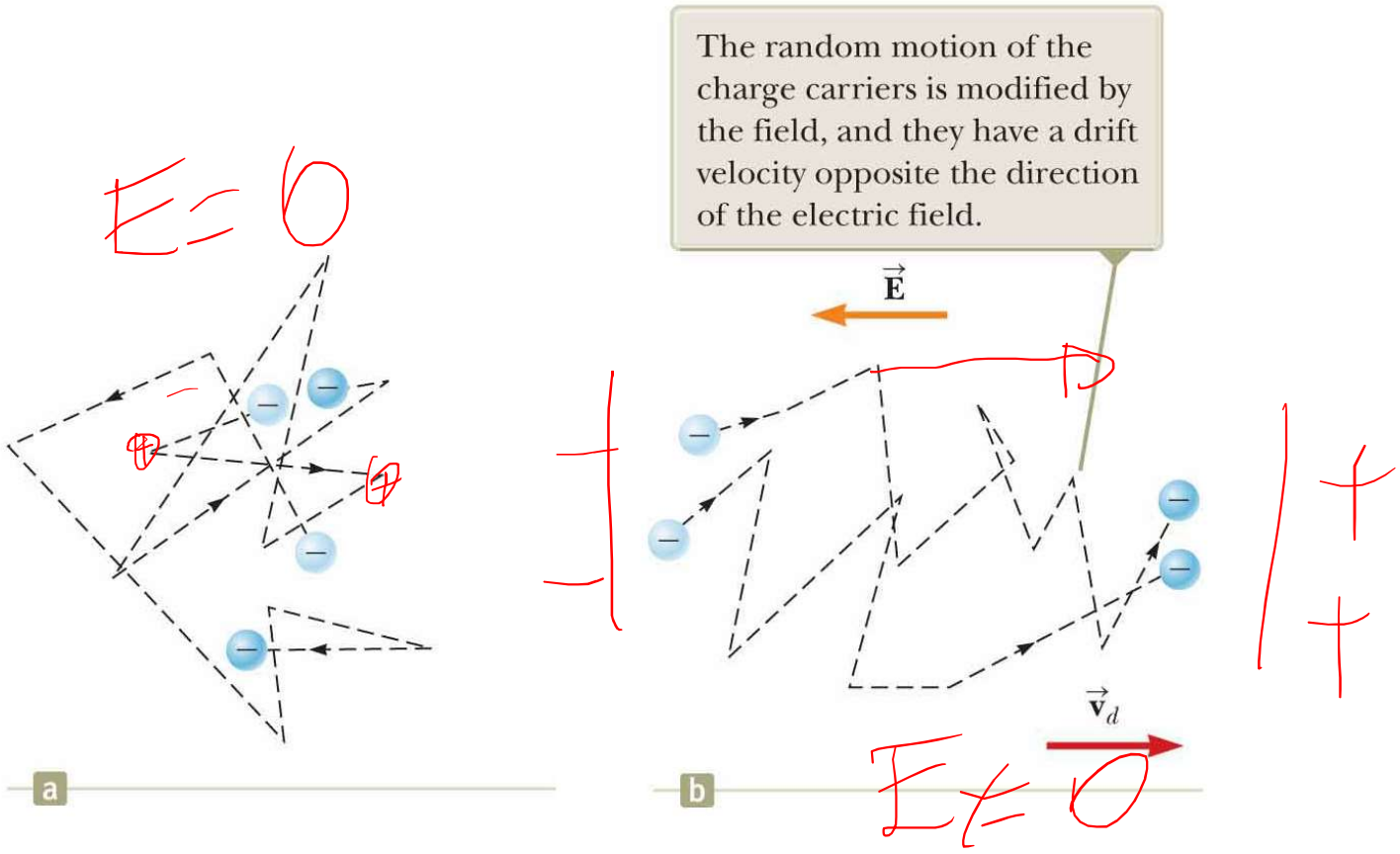
$$I_{\text{avg}} = \frac{\Delta Q}{\Delta t} = nqv_dA$$



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Microscopic Model of Current (2 of 2)

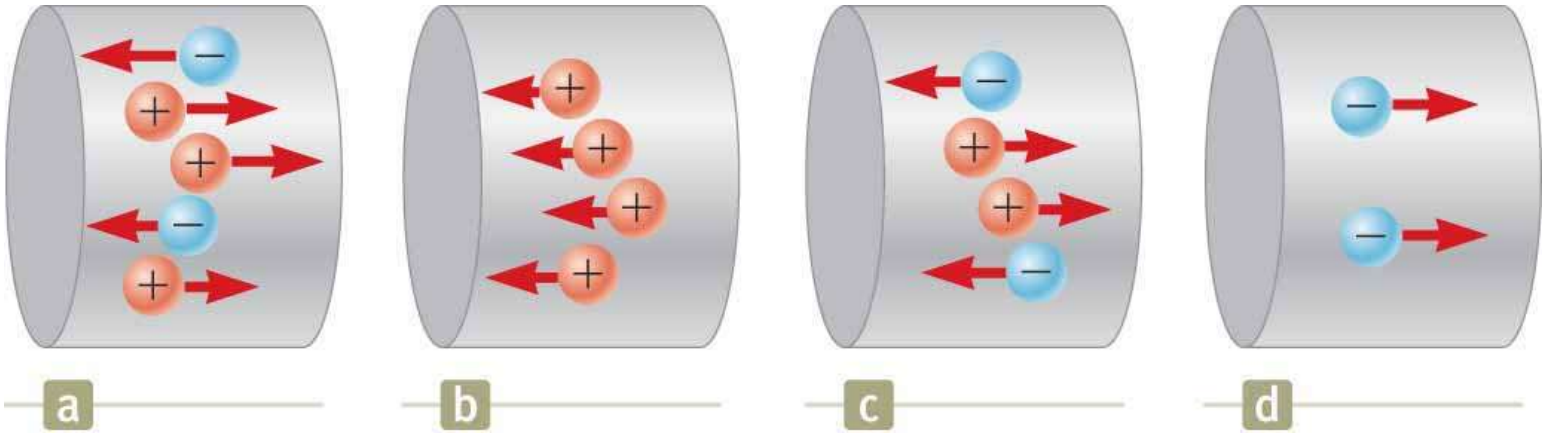


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Quick Quiz 26.1 (1 of 2)

Consider positive and negative charges of equal magnitude moving horizontally through the four regions shown in the figure. Rank the current in these four regions from highest to lowest.

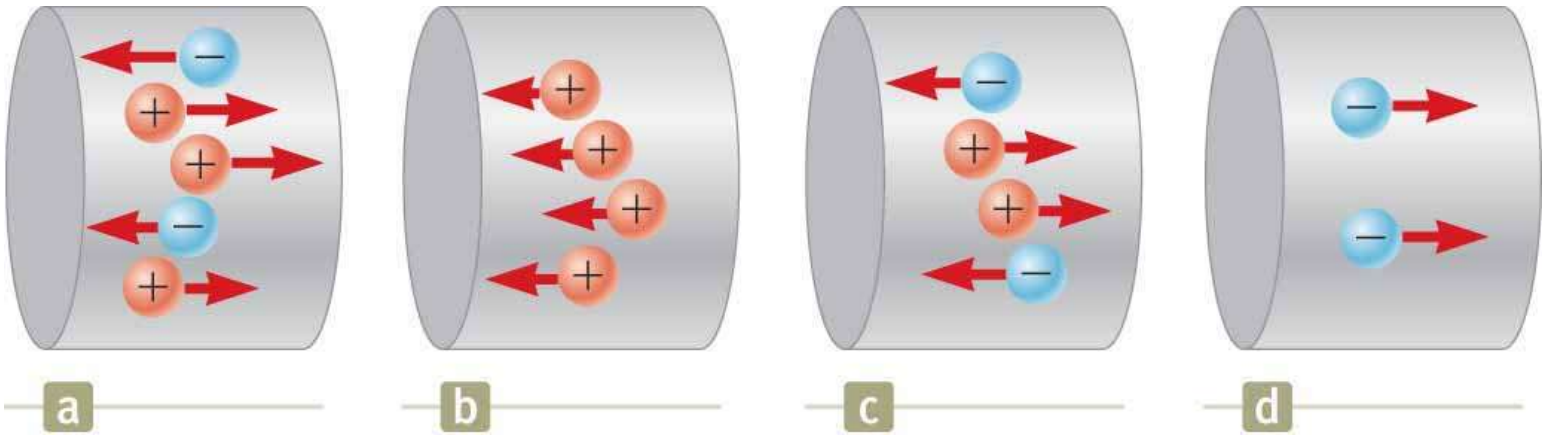


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Quick Quiz 26.1 (2 of 2)

Consider positive and negative charges of equal magnitude moving horizontally through the four regions shown in the figure. Rank the current in these four regions from highest to lowest.



(a) > (b) = (c) > (d)

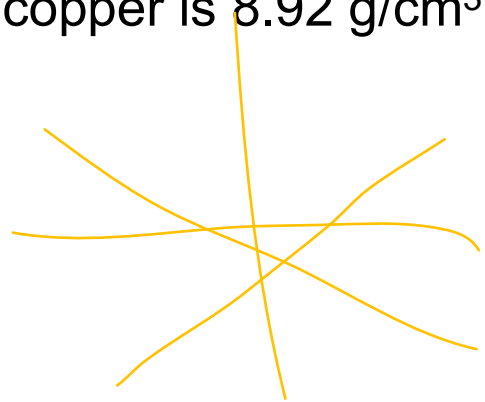
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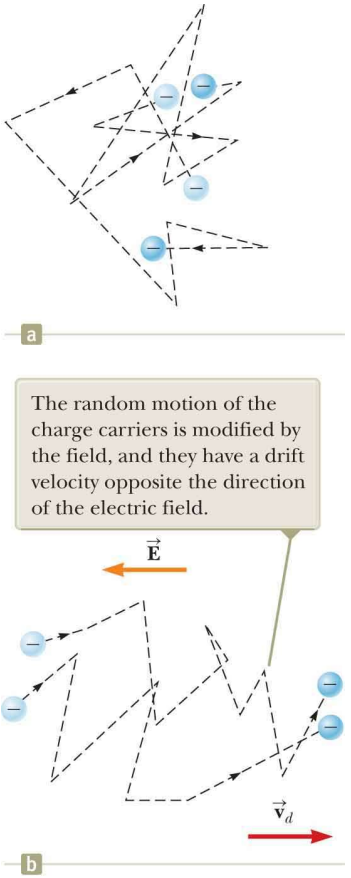
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Example 26.1: Drift Speed in a Copper Wire (1 of 2)

The 12-gauge copper wire in a typical residential building has a cross-sectional area of 3.31×10^{-6} m^2. It carries a constant current of 10.0 A. What is the drift speed of the electrons in the wire? Assume each copper atom contributes one free electron to the current. The density of copper is 8.92 g/cm^3.

Burası çok önemli

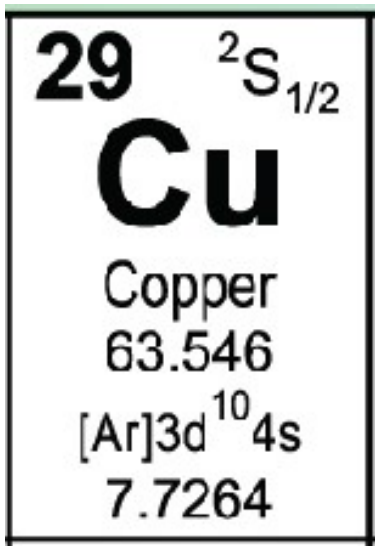




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Example 26.1: Drift Speed in a Copper Wire (2 of 2)



$$V = \frac{M}{\rho}$$

$$n = \frac{N_A}{V} = \frac{N_A \rho}{M}$$

$$v_d = \frac{I_{\text{avg}}}{nqA} = \frac{I}{nqA} = \frac{IM}{qAN_A\rho}$$

$$v_d = \frac{(10.0\ \text{A})(0.0635\ \text{kg/mol})}{(1.60\times10^{-19}\ \text{C})(3.31\times10^{-6}\ \text{m}^2)(6.02\times10^{23}\text{mol}^{-1})(8920\ \text{kg/m}^3)}$$

$$= \boxed{2.23\times10^{-4}\ \text{m/s}}$$

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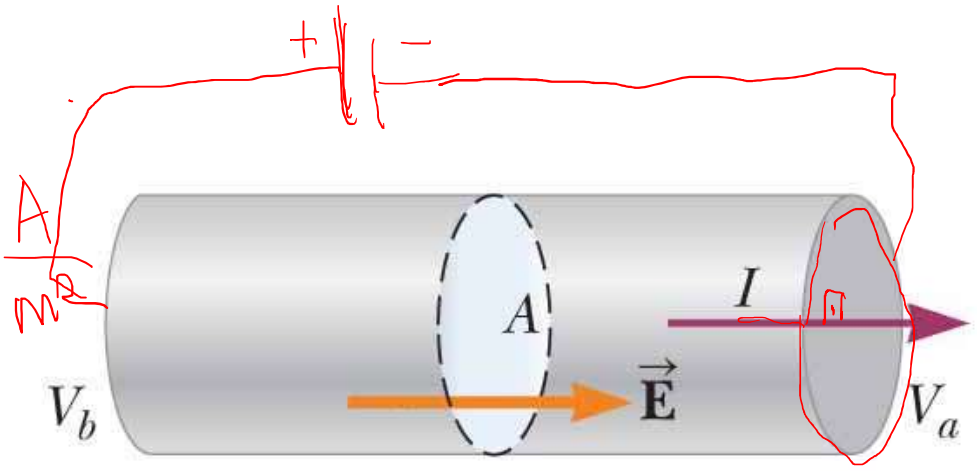
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Resistance (1 of 2)

$$I = nqv_d A$$

$$J \equiv \frac{I}{A} = nqv_d$$

$$J = \sigma E$$



For many materials (including most metals), the ratio of the current density to the electric field is a constant σ that is independent of the electric field producing the current.

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Resistance (2 of 2)

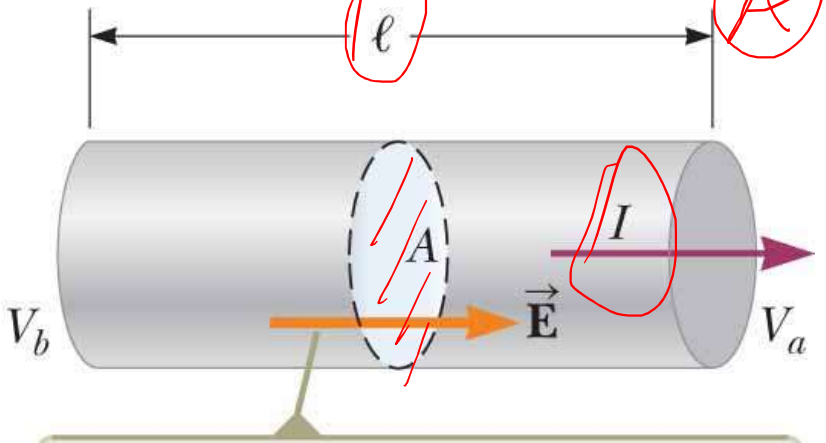
$$\Delta V = E\ell$$

$$\Delta V = \frac{\ell J}{\sigma}$$

$$\Delta V = \left(\frac{\ell}{\sigma A}\right) I = RI$$

$$R \equiv \frac{\Delta V}{I}$$

$$1\,\Omega \equiv 1\ \text{V/A}$$



A potential difference $\Delta V = V_b - V_a$ maintained across the conductor sets up an electric field $\vec{\mathbf{E}}$, and this field produces a current I that is proportional to the potential difference.

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Resistors

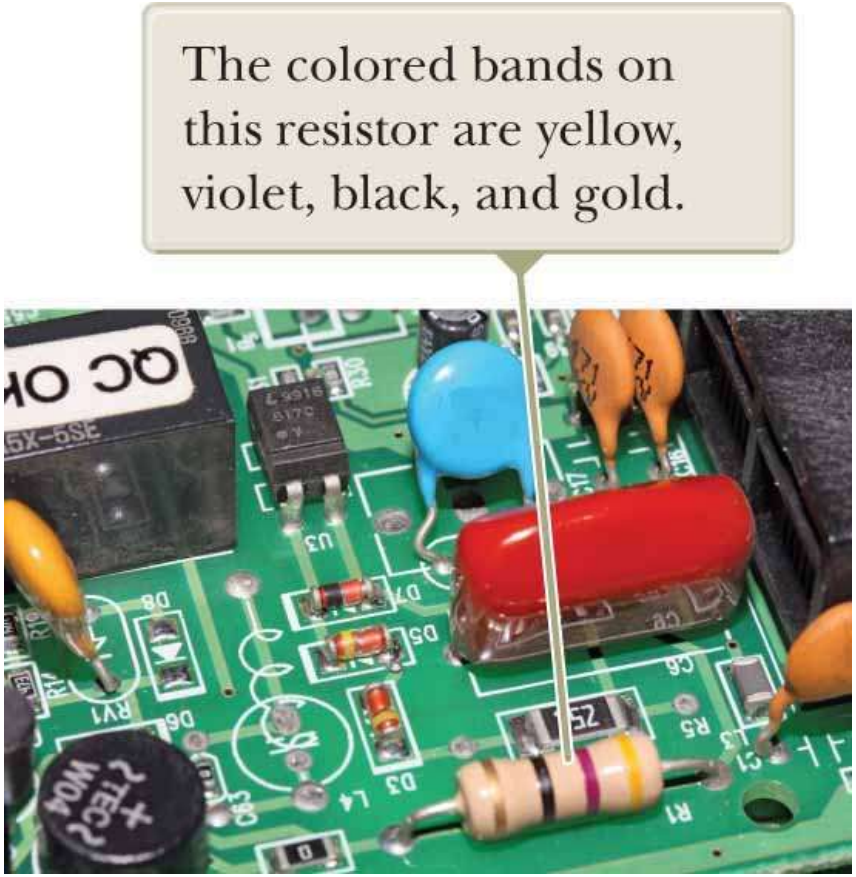


Table 26.1 Color Coding for Resistors

Color	Number	Multiplier	Tolerance
Black	0	1	
Brown	1	10^1	
Red	2	10^2	
Orange	3	10^3	
Yellow	4	10^4	
Green	5	10^5	
Blue	6	10^6	
Violet	7	10^7	
Gray	8	10^8	
White	9	10^9	
Gold		10^{-1}	5%
Silver		10^{-2}	10%
Colorless			20%

$$R = \left(10C_2 + C_1\right) \times 10^{C_3}\ \Omega$$

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$$R = \frac{\ell}{\sigma A} = \rho \frac{\ell}{A}$$

Resistivity

$$\rho = \frac{1}{\sigma}$$

$$R = \rho \frac{\ell}{A}$$

Table 26.2 Resistivities and Temperature Coefficients of Resistivity for Various Materials

Material	Resistivity[a] (Ω · m)	Temperature Coefficient[b] α [(°C)$^{-1}$]
Silver	1.59×10^{-8}	3.8×10^{-3}
Copper	1.7×10^{-8}	3.9×10^{-3}
Gold	2.44×10^{-8}	3.4×10^{-3}
Aluminum	2.82×10^{-8}	3.9×10^{-3}
Tungsten	5.6×10^{-8}	4.5×10^{-3}
Iron	10×10^{-8}	5.0×10^{-3}
Platinum	11×10^{-8}	3.92×10^{-3}
Lead	22×10^{-8}	3.9×10^{-3}
Nichrome[c]	1.00×10^{-6}	0.4×10^{-3}
Carbon	3.5×10^{-5}	-0.5×10^{-3}
Germanium	0.46	-48×10^{-3}
Silicon[d]	2.3×10^{3}	-75×10^{-3}
Glass	10^{10} to 10^{14}	
Hard rubber	$\sim 10^{13}$	
Sulfur	10^{15}	
Quartz (fused)	75×10^{16}	

[a] All values at 20°C. All elements in this table are assumed to be free of impurities.

[b] See Section 26.4.

[c] A nickel–chromium alloy commonly used in heating elements. The resistivity of Nichrome varies with composition and ranges between 1.00×10^{-6} and 1.50×10^{-6} Ω · m.

[d] The resistivity of silicon is very sensitive to purity. The value can be changed by several orders of magnitude when it is doped with other atoms.

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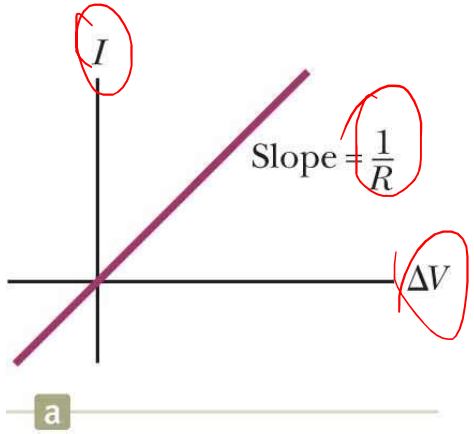
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Current Analogous to Water Flowing in a Pipe

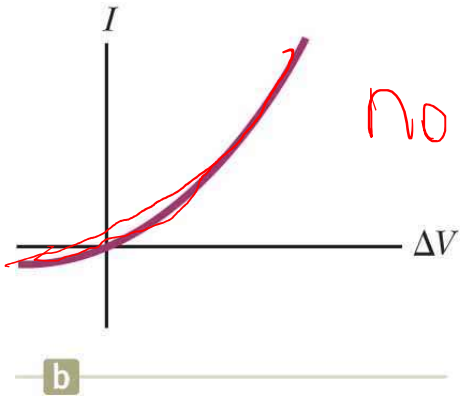


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Ohmic and Nonohmic Materials









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Quick Quiz 26.2 (1 of 2)

A cylindrical wire has a radius r and length ℓ. If both r and ℓ are doubled, does the resistance of the wire

(a) increase

(b) decrease

(c) remain the same

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Quick Quiz 26.2 (2 of 2)

A cylindrical wire has a radius r and length ℓ. If both r and ℓ are doubled, does the resistance of the wire

(a) increase

(b) decrease

(c) remain the same

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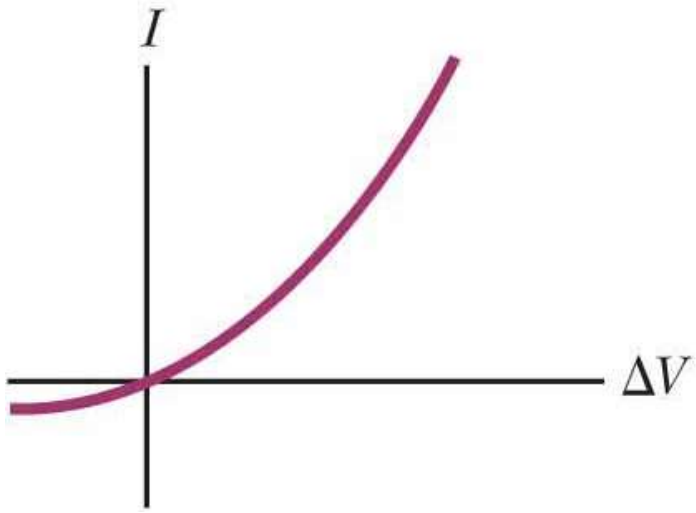
Quick Quiz 26.3 (1 of 2)

In the figure, as the applied voltage increases, does the resistance of the diode

(a) increase

(b) decrease

(c) remain the same



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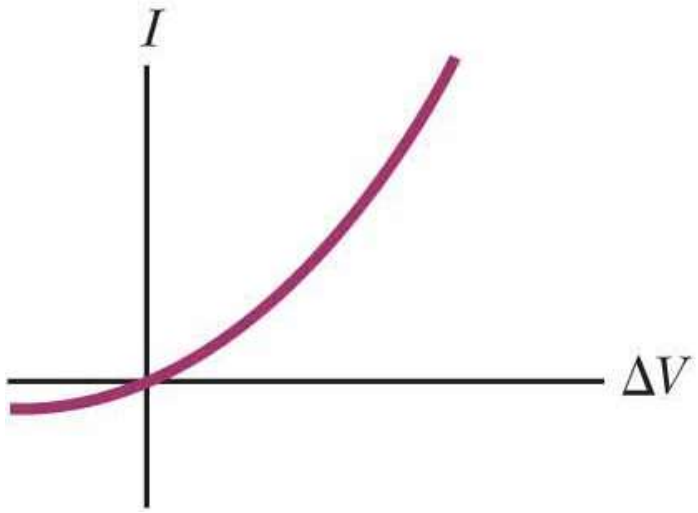
Quick Quiz 26.3 (2 of 2)

In the figure, as the applied voltage increases, does the resistance of the diode

(a) increase

(b) decrease

(c) remain the same



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Example 26.2: The Resistance of Nichrome Wire (1 of 3)

The radius of 22-gauge Nichrome wire is 0.32 mm. (A) Calculate the resistance per unit length of this wire.

$$\frac{R}{\ell} = \frac{\rho}{A} = \frac{\rho}{\pi r^2} = \frac{1.0 \times 10^{-6}\ \Omega \cdot \text{m}}{\pi \left(0.32 \times 10^{-3}\ \text{m}\right)^2} = 3.1\ \Omega/\text{m}$$

Table 26.2 Resistivities and Temperature Coefficients of Resistivity for Various Materials

Material	Resistivity[a] ($\Omega \cdot$ m)	Temperature Coefficient[b] α [(°C)$^{-1}$]
Silver	1.59×10^{-8}	3.8×10^{-3}
Copper	$1.7\,3 \times 10^{-8}$	3.9×10^{-3}
Gold	2.44×10^{-8}	3.4×10^{-3}
Aluminum	2.82×10^{-8}	3.9×10^{-3}
Tungsten	5.6×10^{-8}	4.5×10^{-3}
Iron	10×10^{-8}	5.0×10^{-3}
Platinum	11×10^{-8}	3.92×10^{-3}
Lead	22×10^{-8}	3.9×10^{-3}
Nichrome[c]	1.00×10^{-6}	0.4×10^{-3}
Carbon	3.5×10^{-5}	-0.5×10^{-3}
Germanium	0.46	-48×10^{-3}
Silicon[d]	2.3×10^{3}	-75×10^{-3}
Glass	10^{10} to 10^{14}	
Hard rubber	~10^{13}	
Sulfur	10^{15}	
Quartz (fused)	75×10^{16}	

[a] All values at 20°C. All elements in this table are assumed to be free of impurities.

[b] See Section 26.4.

[c] A nickel–chromium alloy commonly used in heating elements. The resistivity of Nichrome varies with composition and ranges between 1.00×10^{-6} and $1.50 \times 10^{-6}\ \Omega \cdot$ m.

[d] The resistivity of silicon is very sensitive to purity. The value can be changed by several orders of magnitude when it is doped with other atoms.

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Example 26.2: The Resistance of Nichrome Wire (2 of 3)

(B) If a potential difference of 10 V is maintained across a 1.0-m length of the Nichrome wire, what is the current in the wire?

$$I = \frac{\Delta V}{(R/\ell)\ell} = \frac{10\ \text{V}}{(3.1\ \Omega/\text{m})(1.0\ \text{m})} = 3.2\ \text{A}$$

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Example 26.2: The Resistance of Nichrome Wire (3 of 3)

What if the wire were composed of copper instead of Nichrome? How would the values of the resistance per unit length and the current change?

Table 26.2 Resistivities and Temperature Coefficients of Resistivity for Various Materials

Material	Resistivity[a] ($\Omega \cdot m$)	Temperature Coefficient[b] α $[(°C)^{-1}]$
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Platinum	11×10^{-8}	3.92×10^{-3}
Lead	22×10^{-8}	3.9×10^{-3}
Nichrome[c]	1.00×10^{-6}	0.4×10^{-3}
Carbon	3.5×10^{-5}	-0.5×10^{-3}
Germanium	0.46	-48×10^{-3}
Silicon[d]	2.3×10^{3}	-75×10^{-3}
Glass	10^{10} to 10^{14}	
Hard rubber	$\sim 10^{13}$	
Sulfur	10^{15}	
Quartz (fused)	75×10^{16}	

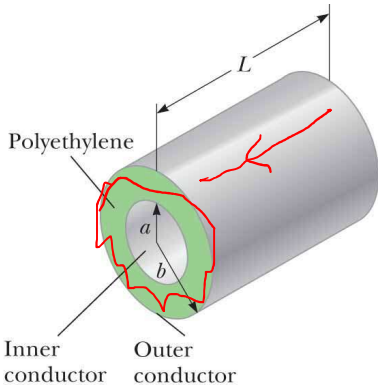
[a] All values at 20°C. All elements in this table are assumed to be free of impurities.
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Example 26.3: The Radial Resistance of a Coaxial Cable (1 of 4)

Coaxial cables are used extensively for cable television and other electronic applications. A coaxial cable consists of two concentric cylindrical conductors. The region between the conductors is completely filled with polyethylene plastic as shown in the figure. Current leakage through the plastic, in the radial direction, is unwanted. (The cable is designed to conduct current along its length, but that is not the current being considered here.) The radius of the inner conductor is $a = 0.500$ cm, the radius of the outer conductor is $b = 1.75$ cm, and the length is $L = 15.0$ cm. The resistivity of the plastic is $1.0 \times 10^{13}\ \Omega \cdot \text{m}$. Calculate the radial resistance of the plastic between the two conductors.



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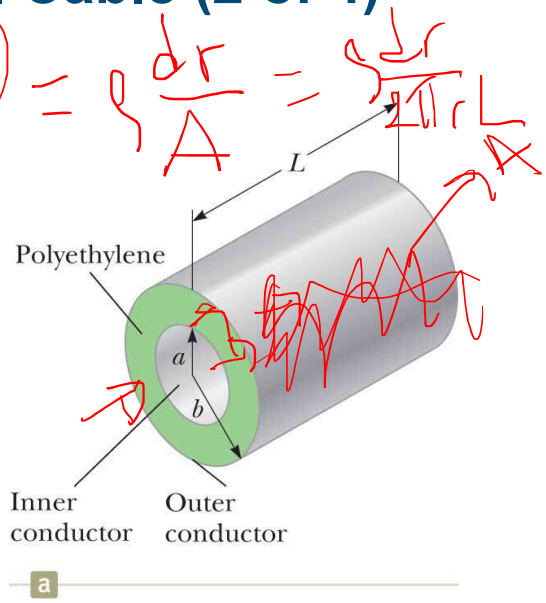
Example 26.3: The Radial Resistance of a Coaxial Cable (2 of 4)

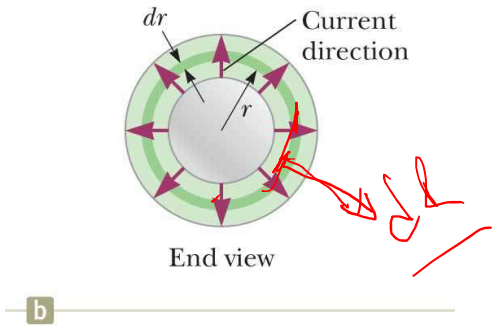
$$dR = \frac{\rho\, dr}{A} = \frac{\rho}{2\pi r L}\, dr$$

$$R = \int dR = \frac{\rho}{2\pi r L}\int_a^b \frac{dr}{r} = \frac{\rho}{2\pi L}\ln\left(\frac{b}{a}\right)$$

$$R = \frac{1.0\times10^{13}\ \Omega\cdot\text{m}}{2\pi\left(0.150\ \text{m}\right)}\ln\left(\frac{1.75\ \text{cm}}{0.500\ \text{cm}}\right)$$

$$= \boxed{1.33\times10^{13}\ \Omega}$$





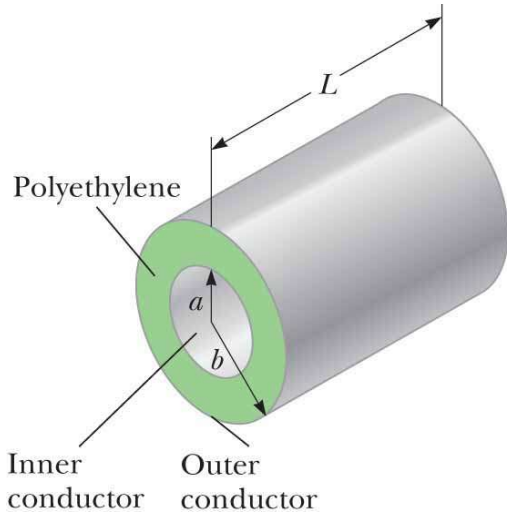
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Example 26.3: The Radial Resistance of a Coaxial Cable (3 of 4)

$$R_{\text{Cu}} = \rho\frac{\ell}{A} = \left(1.7\times10^{-8}\ \Omega\cdot\text{m}\right)\left[\frac{0.150\ \text{m}}{\pi\left(5.00\times10^{-3}\ \text{m}\right)^2}\right]$$

$$= 3.2\times10^{-5}\ \Omega$$



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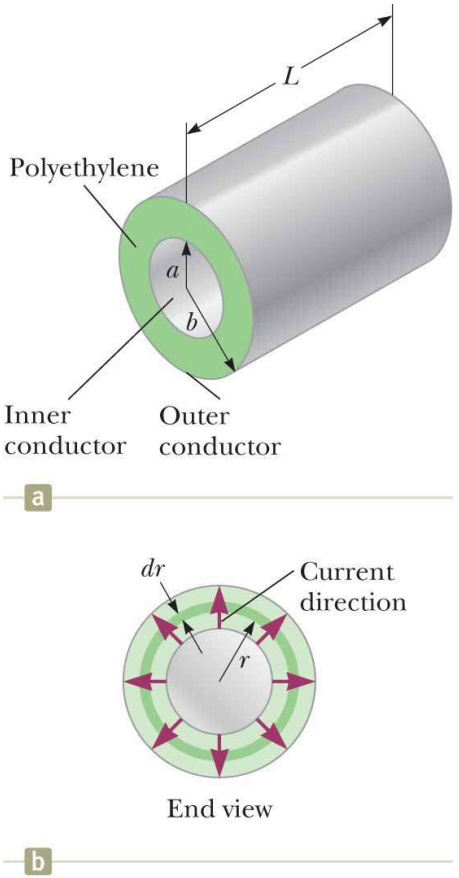
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Example 26.3: The Radial Resistance of a Coaxial Cable (4 of 4)

Suppose the coaxial cable is enlarged to twice the overall diameter with two possible choices: (1) the ratio *b/a* is held fixed, or (2) the difference *b* − *a* is held fixed.

For which choice does the leakage current between the inner and outer conductors increase when the voltage is applied between them?

$$R = \frac{\rho}{2\pi L} \ln\left(\frac{b}{a}\right)$$

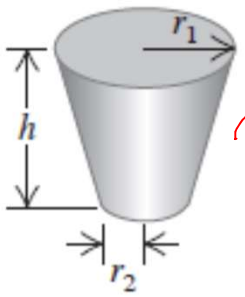


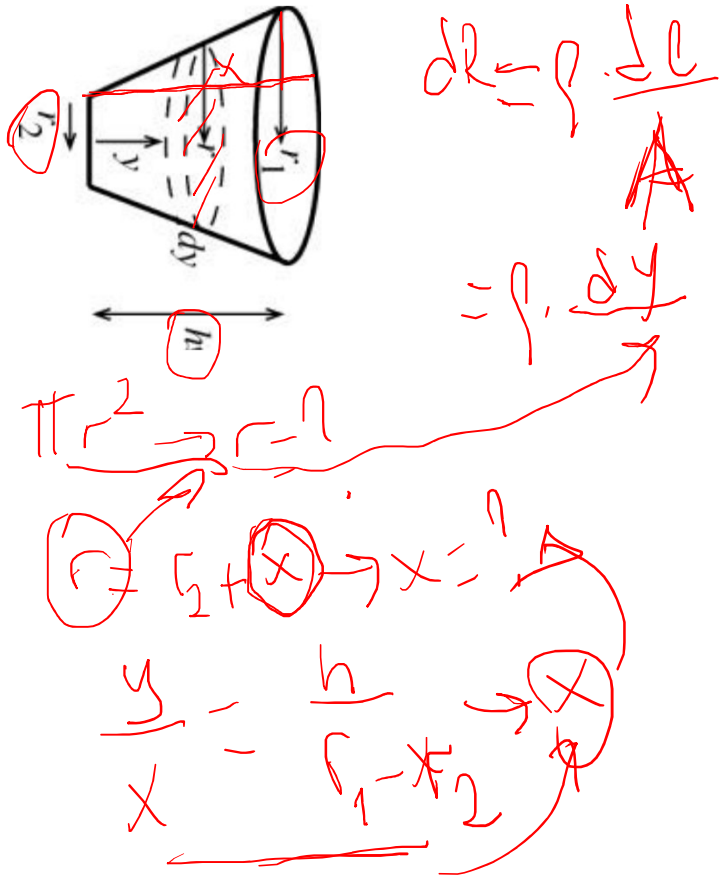
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25.59 • CALC A material of resistivity ρ is formed into a solid, truncated cone of height h and radii r_1 and r_2 at either end (**Fig. P25.59**). (a) Calculate the resistance of the cone between the two flat end faces. (*Hint:* Imagine slicing the cone into very many thin disks, and calculate the resistance of one such disk.) (b) Show that your result agrees with Eq. (25.10) when $r_1 = r_2$.

Figure P25.59





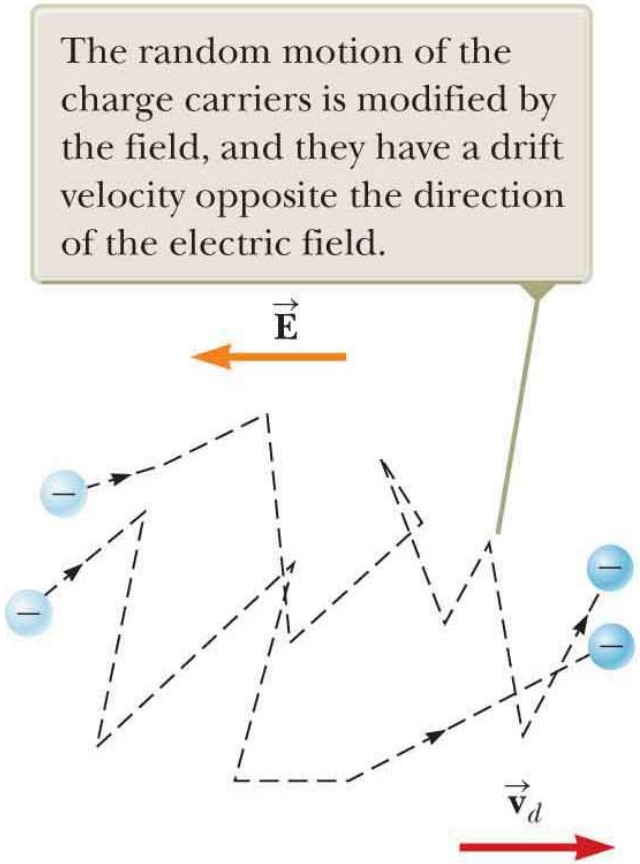
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Drude Model for Electrical Conduction (1 of 3)



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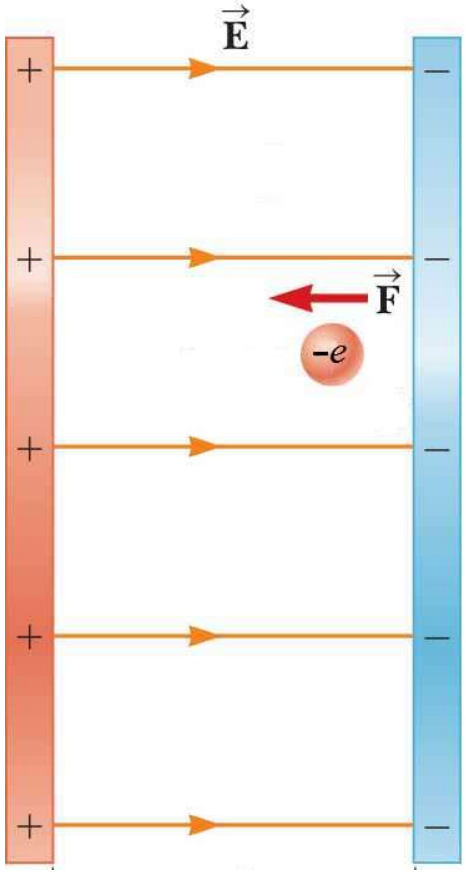
Drude Model for Electrical Conduction (2 of 3)

$$\vec{\mathbf{a}} = \frac{\sum \vec{\mathbf{F}}}{m} = \frac{q\vec{\mathbf{E}}}{m_e}$$

$$\vec{\mathbf{v}}_f = \vec{\mathbf{v}}_i + \vec{\mathbf{a}}t = \vec{\mathbf{v}}_i + \frac{q\vec{\mathbf{E}}}{m_e}t$$

$$\vec{\mathbf{v}}_{f,\text{avg}} = \vec{\mathbf{v}}_d = \frac{q\vec{\mathbf{E}}}{m_e}\tau$$

$$I_{\text{avg}} = nq\left(\frac{qE}{m_e}\tau\right)A = \frac{nq^2\tau A}{m_e}E$$



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Drude Model for Electrical Conduction (3 of 3)

$$J = \frac{nq^2\tau}{m_e}E \qquad\qquad J = \sigma E$$

$$\sigma = \frac{nq^2\tau}{m_e}$$

$$\rho = \frac{1}{\sigma} = \frac{m_e}{nq^2\tau}$$

$$\tau = \frac{\ell_{\text{avg}}}{v_{\text{avg}}}$$

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Resistance and Temperature (1 of 2)

$$\rho = \rho_0 \left[1 + \alpha \left(T - T_0\right)\right]$$

$$\alpha = \frac{\Delta\rho/\rho_0}{\Delta T}$$

$$R = R_0 \left[1 + \alpha \left(T - T_0\right)\right]$$

Table 26.2 Resistivities and Temperature Coefficients of Resistivity for Various Materials

Material	Resistivity[a] (Ω · m)	Temperature Coefficient [b] α [(°C)$^{-1}$]
Silver	1.59×10^{-8}	3.8×10^{-3}
Copper	$1.7\,3 \times 10^{-8}$	3.9×10^{-3}
Gold	2.44×10^{-8}	3.4×10^{-3}
Aluminum	2.82×10^{-8}	3.9×10^{-3}
Tungsten	5.6×10^{-8}	4.5×10^{-3}
Iron	10×10^{-8}	5.0×10^{-3}
Platinum	11×10^{-8}	3.92×10^{-3}
Lead	22×10^{-8}	3.9×10^{-3}
Nichrome[c]	1.00×10^{-6}	0.4×10^{-3}
Carbon	3.5×10^{-5}	-0.5×10^{-3}
Germanium	0.46	-48×10^{-3}
Silicon[d]	2.3×10^{3}	-75×10^{-3}
Glass	10^{10} to 10^{14}	
Hard rubber	~10^{13}	
Sulfur	10^{15}	
Quartz (fused)	75×10^{16}	

[a] All values at 20°C. All elements in this table are assumed to be free of impurities.

[b] See Section 26.4.

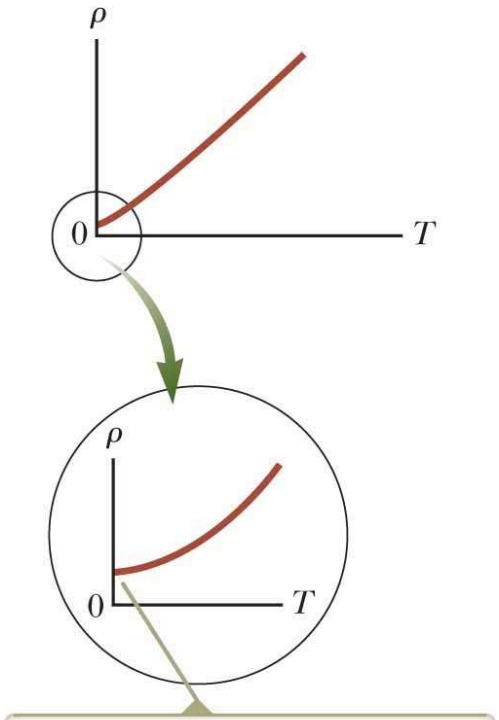
[c] A nickel–chromium alloy commonly used in heating elements. The resistivity of Nichrome varies with composition and ranges between 1.00×10^{-6} and 1.50×10^{-6} Ω · m.

[d] The resistivity of silicon is very sensitive to purity. The value can be changed by several orders of magnitude when it is doped with other atoms.

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Resistance and Temperature (2 of 2)



As T approaches absolute zero, the resistivity approaches a nonzero value.

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Quick Quiz 26.4 (1 of 2)

When does an incandescent lightbulb carry more current,

(a) immediately after it is turned on and the glow of the metal filament is increasing

(b) after it has been on for a few milliseconds and the glow is steady

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Quick Quiz 26.4 (2 of 2)

When does an incandescent lightbulb carry more current,

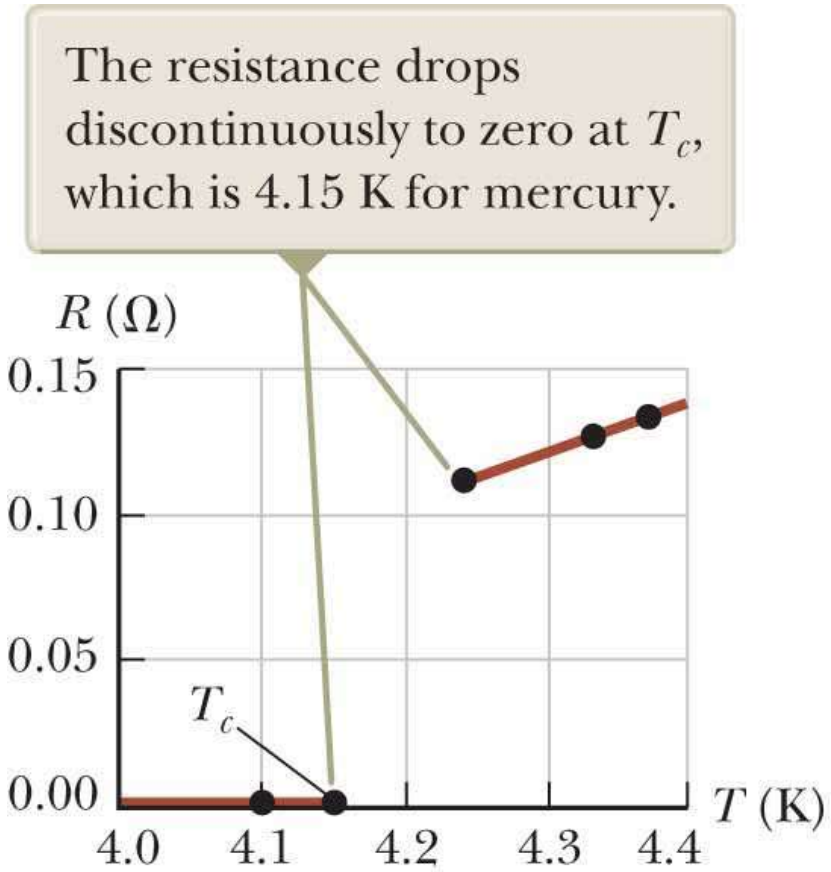
(a) immediately after it is turned on and the glow of the metal filament is increasing

(b) after it has been on for a few milliseconds and the glow is steady

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Superconductors (1 of 2)



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Superconductors (2 of 2)

Table 26.3 Critical Temperatures for Various Superconductors

Material	T_c **(K)**
$HgBa_2Ca_2Cu_3O_8$	134
Tl—Ba—Ca—Cu—O	125
Bi—Sr—Ca—Cu—O	105
$YBa_2Cu_3O_7$	92
Nb_3Ge	23.2
Nb_3Sn	18.05
Nb	9.46
Pb	7.18
Hg	4.15
Sn	3.72
Al	1.19
Zn	0.88

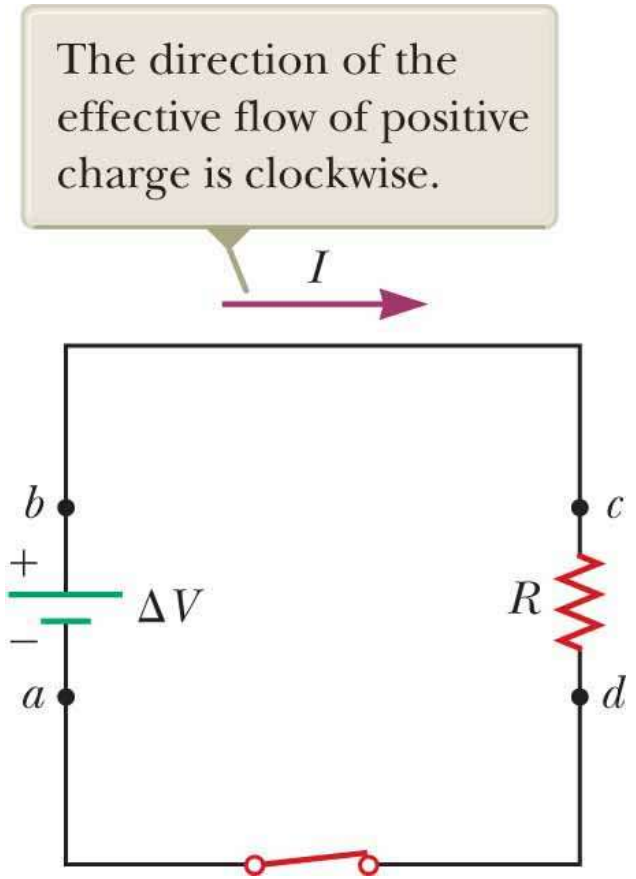
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Electrical Power (1 of 3)

$$\Delta E_{\text{int}} = Q + T_{\text{ET}} + T_{\text{ER}}$$

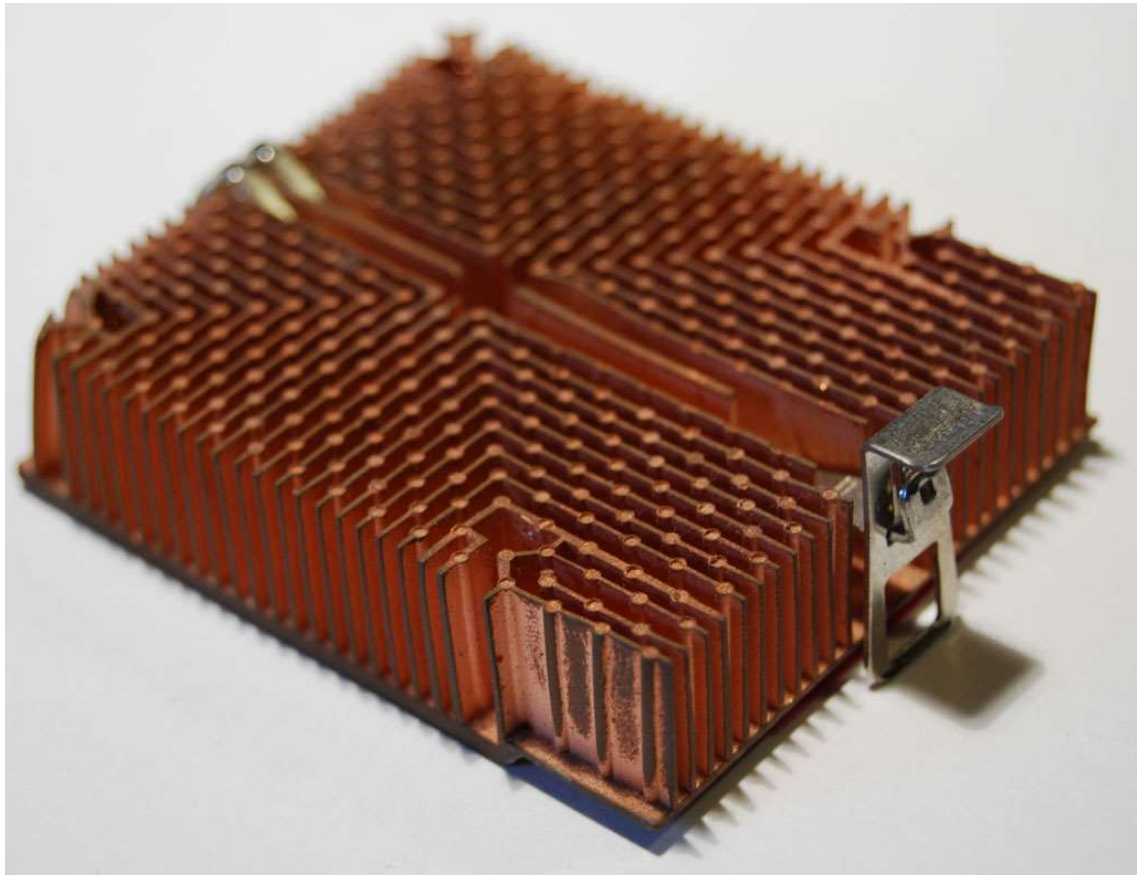
$$\Delta U_{\text{c}} + \Delta E_{\text{int}} = Q + T_{\text{ER}}$$



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Heat Sink



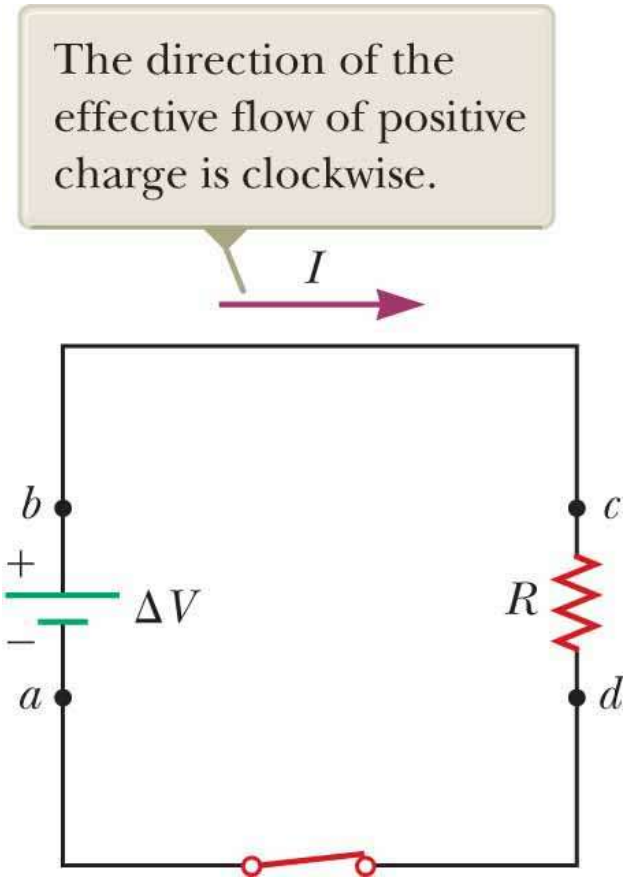
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Electrical Power (2 of 3)

$$\frac{dU_E}{dt} = \frac{d}{dt}(Q\Delta V) = \frac{dQ}{dt}\Delta V = I\Delta V$$

$$P = I\Delta V$$

$$P = I^2 R = \frac{(\Delta V)^2}{R}$$



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Electrical Power (3 of 3)



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Example 26.4: Power in an Electric Heater (1 of 2)

An electric heater is constructed by applying a potential difference of 120 V across a Nichrome wire that has a total resistance of 8.00 W. Find the current carried by the wire and the power rating of the heater.

$$I = \frac{\Delta V}{R} = \frac{120\ \text{V}}{8.00\ \Omega}$$

$$P = I^2 R = (15.0\ \text{A})^2 (8.00\ \Omega) = 1.80 \times 10^3\ \text{W}$$

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Example 26.4: Power in an Electric Heater (2 of 2)

What if the heater were accidentally connected to a 240-V supply? (That is difficult to do because the shape and orientation of the metal contacts in 240-V plugs are different from those in 120-V plugs.) How would that affect the current carried by the heater and the power rating of the heater, assuming the resistance remains constant?

$$R = \frac{\Delta V}{I} \qquad\qquad P = I^2 R$$

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Example 26.5: Linking Electricity and Thermodynamics (1 of 2)

An immersion heater must increase the temperature of 1.50 kg of water from 10.0°C to 50.0°C in 10.0 min while operating at 110 V.

(A) What is the required resistance of the heater?

$$P = \frac{(\Delta V)^2}{R} = \frac{Q}{\Delta t}$$

$$\frac{(\Delta V)^2}{R} = \frac{mc\Delta T}{\Delta t} \rightarrow R = \frac{(\Delta V)^2 \Delta t}{mc\Delta T}$$

$$R = \frac{(110\ \text{V})^2 (600\ \text{s})}{(1.50\ \text{kg})(4186\ \text{J/[kg}\cdot\text{°C]})(50.0\text{°C} - 10.0\text{°C})} = \boxed{28.9\ \Omega}$$

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Example 26.5: Linking Electricity and Thermodynamics (2 of 2)

(B) Estimate the cost of heating the water.

$$T_{\text{ET}} = P\Delta t = \frac{(\Delta V)^2}{R}\Delta t = \frac{(110\ \text{V})^2}{28.9\ \Omega}(10.0\ \text{min})\left(\frac{1\ \text{h}}{60.0\ \text{min}}\right)$$

$$= 69.8\ \text{Wh} = 0.0698\ \text{kWh}$$

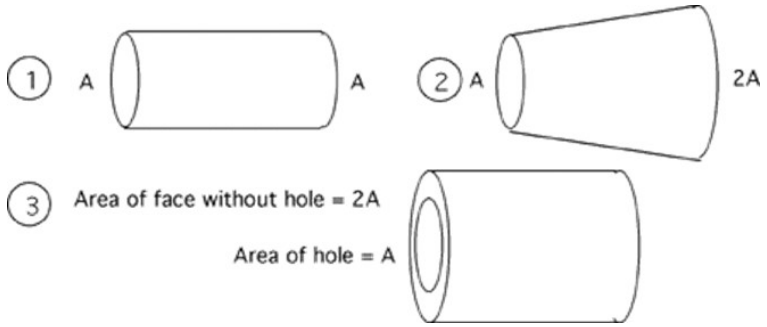
$$\text{Cost} = (0.0698\ \text{kWh})(\$0.11/\text{kWh}) = \$0.008 = \boxed{0.8\ \text{cents}}$$

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Assessing to Learn (1 of 2)

Which object below has the lowest resistance? All three have length L and are made out of the same material.



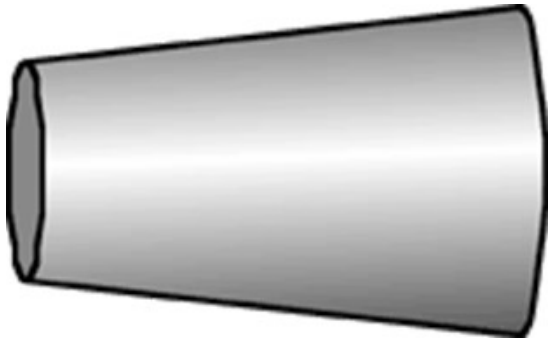
1. #1
2. #2
3. #3
4. Both #1 and #3 have the same, and theirs is less than the resistance of #2.

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Assessing to Learn (2 of 2)

An ohmic conductor is carrying a current. The cross sectional area of the wire changes from one end of the wire to the other. Which of the following quantities vary along the wire?



A. The resistivity
B. The current
C. The current density
D. The electric field

1. A only 2. B only 3. C only 4. D only
5. A and B only 7. A, B, C, and D
6. C and D only 8. None of the above

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